2.4 FULLY AUTOMATIC LIQUID BOTTLES PACKING IN CARTON BOX UNIT WITH ACCESSORIES FOR ERECTION, PACKING, CLOSING AND STRAPPING FOR TRANSPORTATION – REQUIRED NO. 1 SET

Carton Box size for reference

Length - 470mm, Width – 390mm and Height – 270mm

2.4.1. S-Conveyor

Modular belt conveyor should be middle ground between belt conveyors and chain conveyors. A modular belt should consist of individual plastic modules usually linked together with joint rods. This modular belt should be driven by chain sprockets; the base frame of the conveyor should be constructed on standard profile supports. The modular belt should make it possible to accept hot workpieces or workpieces with sharp edges. Furthermore, large conveyor widths should be implemented without high pre-tensioning of the belts. However, in contrast to chain conveyors, this design should less maintenance-intensive and is a technically simpler solution. These systems are frequently used as cooling lines, since the chain modules can be designed to be permeable to air.

Technical Specifications

Machine Type	Automatic
Output per Hour	60 to 65 Bottle/Minute
Direction of Movement	Left to Right
Gear Box (Conveyor + Main)	Suitable Standard Make
Make Motor (Conveyor + Main)	Suitable Standard Make
Working Desk Height	860 To 910 mm according to the requirement
Power supply	240V /15A/50Hz/-1Ph according to the requirement
Power consumption	0.75 Kw according to the requirement
Speed	0-11 mtr/min

Salient Features

- It is one of the simplestconveyors.
- It is used for conveying crates, box, pouches, Bottles &bags.
- It can be moved 90deg, 180deg,270deg.
- It can carry various products up to100kg.
- Size of the conveyor is designed as per therequirement.

- Used in food & pharmaindustries.
- Easy to clean & maintain.
- Easy to replace belt as it is modular.

2.4.2. Case Erector

Case Erectors are used to erect carton which are formed automatically and use tape as sealing materials to seal the bottom flaps of carton. A stack of corrugated blanks should be placed into the machine. The blanks are pulled out of the machine one-by-one and formed into cases. The newly formed cases are sealed at the bottom with tape or glue.

Technical Specifications

Machine Type	Automatic
Output per Hour (No. of cases)	8-10/Minute
Direction of Movement	Left to Right
PLC	Suitable Standard Make
Switchgears	Suitable Standard Make
Sensor	P & F
Power supply	240V /15A/50Hz/-1Ph according to the requirement
Power consumption	0.75 Kw according to the requirement
Speed	0-10 mtr/min
Voltage	1P, 220V, 50Hzaccording to the requirement
Tape	Bopp tape
Tape size	4.8cm–7.2cm(Std machine)
Air supply	Air pipe φ10 mm,≥6Kgf/cm²
Noise	≤75DB
Material of construction	MS powder coated

2.4.3. Roller Conveyor

Roller conveyors are intended for transporting piece goods such as solid boxes or pallets with rigid, flat bases. The drive concept can be selected to suit our particular application. Whether driven by gravity, a tangential chain, a drive roller or a friction drive, all systems are available in straight lines and curves.

Technical Specifications

Machine Type	Automatic
Output per Hour (No. of cases)	8-10/Minute
Direction of Movement	Left to Right
PLC	Suitable Standard Make
Switchgears	Suitable Standard Make
Sensor	P & F
Power supply	240V /15A/50Hz/-1Ph according to the requirement
Power consumption	0.37 Kw according to the requirement
Speed	0-20 mtr/min
Drive roller	SS-304
Structure	MS powder coated
Side guard	Aluminum guide
Foot	Suitable Standard Make
Pneumatic cylinder	Suitable Standard Make

Salient Features

- It is one of the drive roller conveyors.
- It is used in conveying crates & box in 90deg.
- It can carry various products up to 100kg.
- Size of conveyor is designed as per requirement.
- Used in different types of industries.
- Easy to clean & maintain.
- Easy to replace belt as it is modular

2.4.4. Case Tapping Machine

Auto carton sealer/tapping is suitable for carton / case packing, which can be used for single use or production line. It adopts adhesive tape to seal carton/case. It can be finishing the sealing both top and bottom. Also, you can adjust the height and width for different cartons.

Technical Specifications

Machine Type	Automatic
Output per Hour (No. of cases)	8-10/Minute
Direction of Movement	Left to Right
PLC	Suitable Standard Make
Switchgears	Suitable Standard Make
Sensor	P & F
Power supply	240V /15A/50Hz/-1Phaccording to the requirement

Power consumption	0.75 Kwaccording to the requirement
Speed	0-20 mtr/min
Voltage	1P, 220V, 50Hz according to the requirement
Tape	Bopp tape
Tape size	4.8cm–7.2cm(Std machine)
Air supply	Air pipe φ10 mm,≥6Kgf/cm²
Noise	≤75DB

2.4.5. Pick and Place Unit

Typically mounted on a stable stand, pick and place robots are positioned to reach different areas to perform work. They use advanced vision systems to identify, grasp and move objects from one place to another. Matrix of bottle will pick by bottle gripper and place into the boxes as per no. of matrix pick and place can be done.

Technical Specifications

Machine Type	Automatic
Output per Hour (No. of cases)	8-10/Minute
Direction of Movement	Left to Right
PLC	Suitable Standard Make
Switchgears	Suitable Standard Make
Sensor	P & F
Power supply	240V /15A/50Hz/-1Phaccording to the requirement
Power consumption	0.37 Kwaccording to the requirement
Speed	0-20 mtr/min
Drag chain	Suitable Standard Make
Structure	MS powder coated
Linear guide	Suitable Standard Make
Foot	Suitable Standard Make
Pneumatic cylinder	Suitable Standard Make

2.4.6. Idler Roller Conveyor

After case sealing from top of the box filled box will come and accumulate on this gravity roller conveyor. From this point filled box will pick manually

Salient Features

- It is one of the gravity rollerconveyors.
- It is used for conveying crates,box
- It can be moved 90 deg, 180 deg, 270deg.
- It can carry various products up to 100kg.

- Size of conveyor is designed as perrequirement.
- For any industries we can useit.
- Easy to clean & maintain
- Easy is replace roller as it ismodular.

2.4.7. Carton box strapping machine

Auto carton box strapping is suitable for carton / case packing, which can be used for single use or production line. It adopts strap roll to seal carton/case with stoppage clips. It can be finishing the strapping at center and sides of the carton box according to the height and width of the carton box.

Technical Specifications

Machine Type	Automatic
Output per Hour (No. of cases)	2-5/Minute or equal
Direction of Movement	Left to Right
Power supply	100, 110, 220, 230, 240 V., 50 / 60 Hz according to the requirement
Power consumption	400 watt
Strap width	Bopp tape5 -15 mm
Strap speed	1.8sec/strap M/c

Salient Features

- Box strapping machines are widely used to package all kinds and sizes of boxes including corrugated boxes and PP boxes.
- The semi automatic strapping machine with top belt driven, powered table is easy to maintain with easily openable table top. Strapping position adjustable by moving sensor position.
- Box Strapping Machines allow easy loading and unloading of strapping rolls and very carefully and efficiently strap various boxes.
- The number of boxes strapped, depends on the technical specification and size of the strapping machine.
- Some strapping equipment consists of electromagnetic clutch transmission for increased strapping speed and has an adjustable cooling time and strap tension and length for better performance.
- Suitable electrical panel board with voltage switch control like MCB.
- Suitable model of motor and A.C variable frequency drive.

- Required conveyor for manual feeding of carton boxes and also to send strapping machine.

2.5 SEMI-AUTOMATIC LIQUID BOTTLES PACKED CARTON BOX UNIT WITH ACCESSORIES FOR CLOSING AND STRAPPING FOR TRANSPORTATION

2.5.1. Top and Bottom drive type carton sealing machine

Auto carton sealer/tapping is suitable for carton / case packing, which can be used for single use or production line. It adopts adhesive tape to seal carton/case. It can be finishing the sealing both top and bottom. Also, you can adjust the height and width for different cartons.

Carton Box size for reference

Length - 470mm, Width – 390mm and Height – 270mm

Technical Specifications

Machine Type	Automatic
Output per Hour (No. of cases)	2 - 5/Minute or equal
Direction of Movement	Left to Right
Power supply	220Vpl Ac/50Hz according to the requirement
Tape	Bopp tape
Tape size	4.8cm–7.2cm (Std machine)

Salient Features

- Suitable for fast productivity, it is designed for high-speed production with zero maintenance.
- Individual control systemfor pasting the tape.
- Suitable electrical panel board with voltage switch control like MCB.
- Suitable model of motor and A.C variable frequency drive.
- Required conveyor for manual feeding of carton boxes and also to send strapping machine
- **Required – One no.**

2.5.2. Carton box strapping machine

Auto carton box strapping is suitable for carton / case packing, which can be used for single use or production line. It adopts strap roll to seal carton/case with stoppage clips. It can be finishing the strapping at center and sides of the carton box according to the height and width of the carton box.

Technical Specifications

Machine Type	Automatic
Output per Hour (No. of cases)	2-5/Minute or equal
Direction of Movement	Left to Right
Power supply	100, 110, 220, 230, 240 V., 50 / 60 Hz according to the requirement
Power consumption	400 watt
Strap width	Bopp tape5 -15 mm
Strap speed	1.8sec/strap M/c

Salient Features

- Box strapping machines are widely used to package all kinds and sizes of boxes including corrugated boxes and PP boxes.
- The semi automatic strapping machine with top belt driven, powered table is easy to maintain with easily openable table top. Strapping position adjustable by moving sensor position.
- Box Strapping Machines allow easy loading and unloading of strapping rolls and very carefully and efficiently strap various boxes.
- The number of boxes strapped, depends on the technical specification and size of the strapping machine.
- Some strapping equipment consists of electromagnetic clutch transmission for increased strapping speed and has an adjustable cooling time and strap tension and length for better performance.
- Suitable electrical panel board with voltage switch control like MCB.
- Suitable model of motor and A.C variable frequency drive.
- Required conveyor for manual feeding of carton boxes and also to send strapping machine
- **Required – One no.**

2.6 AUTOMATIC BOTTLE PRINTER TO PRINT PLACE OF MANUFACTURE, DATE OF MANUFACTURE, BATCH CODE AND EXPIRY DATE

Automatic bottle printer should print the details of place of manufacture, date of manufacture, batch code and expiry date as per feeding into the bottle by sensor. It should be capable of printing in any text direction and any place of the bottle. It should be in permanent ink.

Technical specifications

Operating Temp	5 to 35 Degree C
Frequency	60 Hz
Humidity	10-80 % RH
Direction	Horizontal
Number of heads	1
Speed	3000-4000 label per hour
Printing speed	Intermittent 100 mm/s,150 mm/s,200 mm/s, 225 mm/s, Continuous up to 450 mm/s
Voltage	90 V to 260 V,47 Hz to 63 Hz
Resolution	8 dots per mm
Print area	30mm to 50mm
Message Storage Capability	250
Maximum Lines of Print	5, 32pixel
Print Fonts	5, 7, 9, 12, 16, 19, 25, 32
Print Head Cleaning	Automatic
Printer ink	One set with machine and additional one set for further use

Electricals

- MCC Panel board With power cables and control cables
- Panel board, switches and cables – make and wire gauge should be mentioned.
- Electrification Material like Cables, Cable trays, Clamps, Earthing, G.I strips necessary for power wiring from Panel to respective Motors.
- Earthing, consisting of earth pits, running of earth wire etc for MCC panel and individual motor

Erection, Commissioning and trial run

- Complete erection of machines, drives, structure, Chequered plates and electrical till MCCpanel.
- Commissioning of total system toproduction.
- Minimum two months training to locally appointed staff for operating liquid production plant. Training should be free ofcost.

- Annual Maintenance charges should be free at least for first one year from the date of commissioning of feed plant.
- Any service work should be attended within 48hours.
- Guarantee and Warranty period should bespecified.
- Spare parts with price list should be submitted.

Structure

- Suitable structural design and quantities should be designed for standard production plant.
- Stability certificate should be provided after erection.
- Structural drawing for erection of machineries including platforms and ladder should be given.

Complete monitoring of plant by computerized Monitoring system

- PLC Panel.
- PLC control panel to control the above batching plant.
- Batch report generation and storage.
- Connectivity to PC for data transfer for printing.
- Desktop Computer.
- ***Rate should be quoted separately for complete monitoring system.***